Regular low-object absorption, far square completion, and progressive boundary

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1 Purpose and scope

R-079 gave the exact expectation-level decomposition of the complete canonical safe packet. It left four items in its immediate work list: the conditional low-current block, the complete low safe endpoint, a far-root production estimate, and a near-root production estimate. This note closes the first two in the declared regular control class and makes the last two strictly more precise.

There is also a necessary scope correction. The fixed-cutoff graph recovery of R-075 is a closure theorem for the mutually orthogonal, whole-shell, strict-past one-shot class used throughout R-066–R-079. It is not a density theorem for every Boue–Dupuis progressive covariance-flow control. In particular it does not create within-shell temporal feedback, repeated use of the same spatial range, or auxiliary randomness. A lower bound on that restricted class cannot by itself lower-bound the infimum over the larger progressive class. A full-progressive/revisit extension is therefore a separate load-bearing theorem between the regular packet estimate and the canonical one-use inequality.

The positive result of this note is consequently scoped as follows. Fix the terminal cutoff J, the positive production floor, the common even regulator, the A1 production frame, and a sufficiently large but fixed j_0. Controls are regular, mutually orthogonal, one-shot whole-shell controls with h_j measurable with respect to $\mathcal{F}_{j-1}$. No later control revisits a low spatial range. All constants in the two low estimates are uniform in J but may depend on j_0, the floor, and the declared budgets.

Within this scope:

1. the R-079 conditional low current is below arbitrary terminal-sextic budget plus a constant;
2. the distinct complete low safe endpoint obeys the same type of bound;
3. orthogonal far-shell square completion absorbs both terminal-feedback channels and leaves one localized predictable base-current tail;
4. the residual near payload is predictable, but a future-adapted high–high-to-low coefficient branch remains;
5. target-variable heat projection and the weighted Cameron–Martin square function do not automatically supply a probability-root/spatial- shell gap;
6. general progressive/revisit extension remains unproved and cannot be inferred from R-075.

No regular near/far production bound, full progressive extension, controlled-shell one-use theorem, Nelson estimate, measure theorem, or Sector-A closure is asserted. The tier stays T4.

2 Authority and the two low objects

Put $\ell = j_0 - 1$, $A^* = A^{(J)}$, $A^0 = A^{(\ell)}$, and $Z^* = U + A^*$. R-079 places the complete current in the Hilbert space carrying the spatial integral, generator and direction sums, and the positive Q_{II} metric:

$$\mathcal{J}(H)_{\alpha i} = Q_{\mathrm{II}}^{1/2} M_\alpha(U+H)^T (D_i U + D_i H). \tag{2.1}$$

Let $P_\ell = \mathbb{E}[\,\cdot \mid \mathcal{F}_\ell]$ and let $B(z) = \sum_\alpha M_\alpha(z) Q_{\mathrm{II}} M_\alpha(z)^T$. The first low object in the exact R-079 identity is

$$\begin{aligned}\mathfrak{L}_\ell &= \frac{1}{2}\mathbb{E}(\|P_\ell \mathcal{J}(A^*)\|_{\mathcal{H}}^2 - \|P_\ell \mathcal{J}(A^0)\|_{\mathcal{H}}^2) \\ &\quad - \frac{1}{2}\mathbb{E}\sum_i \int \{B(Z^*) - B(U+A^0)\} : \Gamma_{\ell,i}\,\mathrm{d}x.\end{aligned} \tag{2.2}$$

The second low object is the complete low safe endpoint

$$\mathfrak{B}_{\mathrm{low}} = \Delta V_J^{\mathrm{ren}}(U; A^0) - N_{3,\mathrm{nr}}(U; A^0) - \widehat{T}_{\le}(U; A^0). \tag{2.3}$$

Equation (2.3), not merely its R-066 finite-low component, is what appears in the canonical safe packet. Equations (2.2) and (2.3) are different objects and will not be conflated.

The production frame bounds used below are the R-066/R-069 estimates

$$|M(z)| \le C_M |z|, \qquad |M(z+a) - M(z)| \le C_{\mathrm{Lip}} |a|, \tag{2.4}$$

with $C_M = \sqrt{20}$ sufficient, and hence

$$0 \le B(z) \le C_Q |z|^2 I. \tag{2.5}$$

The future value covariance is uniformly bounded at fixed positive floor. The heat-averaged frame therefore satisfies

$$|\overline{M}_\ell(z)| \le C_e(|z| + \sigma_>). \tag{2.6}$$

3 Closure of both regular low objects

3.1 Conditional low current

Theorem 3.1 (regular conditional-low absorption). For every $\zeta > 0$, under the scope of Section 1,

$$\boxed{\mathfrak{L}_\ell \ge -\zeta \mathbb{E}\|Z^*\|_6^6 - C_{\zeta,e,j_0}.} \tag{3.1}$$

The constant is uniform in the terminal cutoff J.

Proof. Both B and each $\Gamma_{\ell,i}$ are positive semidefinite. In (2.2) drop the positive $\|P_\ell \mathcal{J}(A^*)\|^2$ term and the positive $B(U+A^0) : \Gamma_\ell$ term. By (2.5),

$$\mathfrak{L}_\ell \ge -\frac{1}{2}\mathbb{E}\|P_\ell \mathcal{J}(A^0)\|_{\mathcal{H}}^2 - C_{e,j_0}\mathbb{E}\|Z^*\|_2^2. \tag{3.2}$$

The no-revisit property is now used exactly once. With $Z_\ell = P_{<j_0} Z^*$, the predictable translation version of the R-079 heat identity gives

$$P_\ell \mathcal{J}(A^0) = Q_{\mathrm{II}}^{1/2} \overline{M}_\ell(Z_\ell)^T D Z_\ell. \tag{3.3}$$

Because Z_ℓ has fixed spatial frequency, Bernstein and (2.6) imply

$$\|P_\ell \mathcal{J}(A^0)\|_2^2 \le C_{e,j_0}\{1 + \|Z_\ell\|_6^4\} \le C_{e,j_0}\{1 + \|Z^*\|_6^4\}. \tag{3.4}$$

The last projection constant is fixed once j_0 is fixed. The scalar Young inequalities

$$x^4 \le \zeta x^6 + \frac{4}{27\zeta^2}, \qquad x^2 \le \zeta x^6 + \frac{2}{3\sqrt{3\zeta}} \tag{3.5}$$

permit the coefficients in (3.2)–(3.4) to be absorbed into an arbitrarily prescribed terminal-sextic budget after rescaling ζ. This proves (3.1). No control-energy allocation is required. ☐

3.2 Complete low safe endpoint

Theorem 3.2 (regular complete-low absorption). For every $\zeta > 0$, under the same scope,

$$\boxed{\mathbb{E}\mathfrak{B}_{\mathrm{low}} \geq -\zeta\mathbb{E}\|Z^*\|_6^6 - C_{\zeta,e,j_0}.} \tag{3.6}$$

Proof. R-066 proves the finite-low estimate for the complete translated renormalised-energy block before the R-078 paid subtraction:

$$\mathbb{E}\Delta V_J^{\mathrm{ren}}(U; A^0) \geq -\zeta_1\mathbb{E}\|Z^*\|_6^6 - C_{\zeta_1,e,j_0}. \tag{3.7}$$

R-078 pays $N_{3,\mathrm{nr}} + \widehat{T}_{\leq}$ by arbitrary internal energy and sextic budgets and an R-050 random remainder with moment $30/7$. At A^0 every control factor lies in the fixed finite-dimensional low range. Moreover the no-revisit identity writes

$$A^0 = P_{<j_0}Z^* - P_{<j_0}U. \tag{3.8}$$

Fixed-low Bernstein, the bounded low Gaussian moments of U, and (3.5) therefore turn both internal X_0 and Y_0 payments into $\zeta_2\mathbb{E}\|Z^*\|_6^6 + C_{\zeta_2,e,j_0}$. Choose the internal budgets so that $\zeta_1 + \zeta_2 \leq \zeta$. Combining with (3.7) gives (3.6). □

The two theorems remove the low range as a UV obstruction in the regular one-shot class. They do not say that the two objects can be bounded separately after a later control revisits and cancels the low range; Section 6 shows that statement is false.

4 Far region: exact square completion and the true successor

Let Π_m denote orthogonal spatial Fourier-shell projections. In the R-079 future block put

$$b_{j,m} = \Pi_m d_j\mathcal{J}_j, \qquad q_{j,m} = \Pi_m i_j, \tag{4.1}$$

where i_j is the complete terminal future-feedback current. It includes both

$$d_j\{(\Delta_j^{\mathrm{f}}M)^T(DU + DA^{(j)})\}, \qquad d_j\{M(U + A^*)^T DR_j^{\mathrm{f}}\}. \tag{4.2}$$

For every coordinate in the positive Q_{II} Hilbert metric,

$$\langle b_{j,m}, q_{j,m}\rangle_{Q_{\mathrm{II}}} + \frac{1}{2}\|q_{j,m}\|_{Q_{\mathrm{II}}}^2 = -\frac{1}{2}\|b_{j,m}\|_{Q_{\mathrm{II}}}^2 + \frac{1}{2}\|b_{j,m} + q_{j,m}\|_{Q_{\mathrm{II}}}^2. \tag{4.3}$$

Consequently, on the far region $m \geq j + C$, both feedback channels can be retained and absorbed by their positive square. The entire new far loss is the localized predictable base-current tail

$$S_C = E\sum_j \sum_{m\geq j+C} \|\Pi_m d_j\mathcal{J}_j\|_{Q_{\mathrm{II}}}^2. \tag{4.4}$$

This is stronger and cleaner than estimating coefficient feedback and derivative feedback separately. It is also only a reduction: no cutoff-uniform bound for (4.4) is currently available.

For the $\mathcal{F}_{j-1}$-predictable partial control let $Z_{j-1} = U_{j-1} + A^{(j)}$ and let g_j be the fresh Gaussian shell. The R-079 heat projection expands the base increment exactly into

$$\begin{aligned} d_j\mathcal{J}_j = {} & \overline{M}_j(Z_{j-1} + g_j)^T Dg_j \\ & + \{\overline{M}_j(Z_{j-1} + g_j) - \overline{M}_j(Z_{j-1})\}^T DZ_{j-1} \\ & + \{\overline{M}_j - \overline{M}_{j-1}\}(Z_{j-1})^T DZ_{j-1}. \end{aligned} \tag{4.5}$$

These are derivative injection, value innovation, and the heat compensator. The scale data are

$$\mathbb{E}\|g_j\|_2^2 \lesssim 2^{-j}, \qquad \mathbb{E}\|Dg_j\|_2^2 \lesssim 2^{j}, \qquad \|\Sigma_j\| \lesssim 2^{-j}. \tag{4.6}$$

Thus the heat compensator is the easier term. The missing theorem is the spatial root-output tail for the derivative-injection/value-innovation pair under a random predictable translation.

The covariance trace is not a new far obstruction. It must remain in the R-070 Abel-summed form, for which

$$|\mathcal{P}_J| \lesssim \sum_k \left[2^{-k}\|U_J + A_k\|_2\|h_k\| + 2^{-3k}\|h_k\|^2\right]. \tag{4.7}$$

Dyadic Hardy and Young allocate (4.7) below arbitrary global budgets. An artificial spatial-m trace split would discard this proved cancellation.

A sufficient far theorem would give, for some $\gamma > 0$, the localized gain

$$\|\Pi_{\geq j+C}\mathbb{K}_{\gamma,j}(A)\|_{H^{-3/5}} \leq 2^{-\gamma C} R_\gamma \|A\|_{H^{2/5}}, \qquad R_\gamma \in \bigcap_{p<\infty} L^p. \tag{4.8}$$

The R-078/R-079 ledger would then be

$$2^{-\gamma C} R_\gamma X^{1/2-\gamma/4} Y^{1/2+\gamma/12}. \tag{4.9}$$

Its Young slack is $\gamma/6$ and the required random moment is $6/\gamma$. At $\gamma = 3/10$ the powers are $17/40, 21/40$, the slack is $1/20$, and the random moment is 20. R-063 motivates a possible sewing range but does not prove (4.8) for random predictable translations.

5 Why heat projection does not manufacture spatial gap

The semigroup in (4.5) acts on the six-real target variable of M. It is not a Fourier multiplier in x. A simple exact model already separates the two operations. For $H(u) = u^3$,

$$P_\sigma H(u) = u^3 + 3\sigma u. \tag{5.1}$$

If $u(x) = r\cos(Nx)$, the $3N$ Fourier coefficient of the first term is $r^3/4$ before and after target heat averaging. Target heat changes the lower harmonic but does not spatially bandlimit the nonlinear composition. The production rational frame has the same general composition issue; a production paracomposition estimate, not the word "heat", must prove any spatial tail.

There is also an admissible root/shell saturation fixture. Take $j < k$, a centered bounded function $\phi(\xi_j)$, and the later one-shot control

$$h_k = \phi(\xi_j)e_k, \qquad a_k = K_k h_k. \tag{5.2}$$

Then h_k is $\mathcal{F}_{k-1}$-measurable and $d_j a_k = a_k$. With the normalized order-minus-two multiplier,

$$2^{4k}\mathbb{E}\|d_j a_k\|_2^2 = 1 \tag{5.3}$$

for every gap $k - j$. At a predictable background with $M(z_0)^T e_k \neq 0$, derivative feedback likewise has

$$2^k\|d_j D a_k\|_2 \asymp 1. \tag{5.4}$$

Thus no extra $2^{-\theta(k-j)}$, $\theta > 0$, follows from the weighted Cameron–Martin identity or base-current heat identity alone. Equations (5.1)–(5.4) are method boundaries, not counterexamples to the complete production packet or to a production-specific signed far estimate.

6 Near region: predictable payload, hidden coefficient

Use the corrected R-078 transport coordinate

$$N_3 = \int_0^1 (1-t) \left\langle \mathcal{J}, Q[\widehat{K}_t[A,A]]^T DA \right\rangle \, \mathrm{d}t, \qquad \widehat{K}_t = D^2M(U+tA) - D^2M(U). \tag{6.1}$$

Let r be the largest explicit input shell of the A^2DA payload and use the R-077/R-078 split $\widehat{T}_>$ with $m > r + L$. Choose $L \geq C$. In the near-root region $|m-j| \leq C$, any explicit future control shell $k > j$ would force

$$r \geq k > j, \qquad m \leq j + C \leq r + L, \tag{6.2}$$

which places that monomial in the already paid $\widehat{T}_\leq$ branch. Hence the explicit residual payload is $\mathcal{F}_{j-1}$-measurable.

This does not close the near term. The coefficient $\Delta_m \widehat{K}_t$ can hide later adapted high–high-to-low control inputs inside the nonlinear composition. R-063 treats deterministic translates and their balanced jets, not this future-random coefficient. A largest coefficient-input shell split gives the honest successor:

- $p \leq j + O(1)$ is the candidate conditional R-063/R-050 jet branch;
- $p \gg j$ is a genuine pair-high Cameron–Martin harvest or production-specific signed-cancellation branch.

If the first branch could be reduced to a truly predictable translation, the balanced-jet regularity would permit a candidate

$$0 < \gamma < \alpha - \frac{2}{5} - \kappa < \frac{1}{10}. \tag{6.3}$$

For example $\gamma = 1/20$ gives

$$X^{39/80} Y^{121/240} R_{1/20}, \qquad 1 - \frac{39}{80} - \frac{121}{240} = \frac{1}{120}, \qquad p_R = 120. \tag{6.4}$$

This is a viable conditional ledger, not an established adapted operator bound. Merely restricting $|m-j| \leq C$ corresponds to $\gamma = 0$, giving $X^{1/2}Y^{1/2}$ and zero Young slack. Bounded shell width removes counting but does not create absorption slack.

Universal pointwise/rootwise positivity is unavailable even for the actual production frame. For the exact fixture $z = e_1 + e_3$, $a = e_1 - e_3$, $y = e_3$, and $A = a\mathbf{1}_{\{|\xi|>r\}}$ with $DA = 0$, the complete covariance-normal root contribution satisfies

$$\mathbb{E}\{W(A) - W(0)\} = -\frac{3}{80P(2+e)^2} \, r\varphi(r) < 0. \tag{6.5}$$

The terminal square, trace, and complete forest are present; derivative feedback and the paid N_3 vanish. Equation (6.5) refutes a universal PSD/Ward/pointwise Schur claim. It does not refute a frequency-local, budgeted production near estimate.

7 Progressive revisits: a separate theorem is mandatory

The low proofs use $Z_\ell = P_{<j_0} Z^*$, which is true for the declared orthogonal no-revisit class. It fails for a general progressive control that later reuses the same spatial range. The failure is not cosmetic.

Choose a non-null low production mode f with $D(f^T S_r f) \neq 0$. First use a low control $A^0 = tf$ and later revisit the same range with $-tf$, so $A^* = 0$. The production current contains

$$J_{r,i}(U + tf) = t^2 D_i(f^T S_r f) + O(t). \tag{7.1}$$

The positive Q_{II} current metric has $c_0 = 3/(250P) > 0$, and conditional heat projection preserves the deterministic leading term. Therefore

$$\|P_\ell \mathcal{J}(A^0)\|^2 = ct^4 + O(t^3), \qquad c > 0, \tag{7.2}$$

while the covariance trace and the two control costs are $O(t^2)$. Since the final field is independent of t, (2.2) gives

$$\mathfrak{L}_\ell = -\frac{c}{2}t^4 + O(t^3). \tag{7.3}$$

No estimate of $\mathfrak{L}_\ell$ separately by an arbitrary small multiple of progressive control energy plus final sextic charge can hold on this larger class. The later current blocks may cancel (7.3), so this is not a counterexample to the complete action or to canonical one-use.

The logical order is equally decisive. If $\mathcal{A}_{\mathrm{reg}} \subset \mathcal{A}_{\mathrm{prog}}$, then

$$\inf_{\mathcal{A}_{\mathrm{prog}}} F \le \inf_{\mathcal{A}_{\mathrm{reg}}} F. \tag{7.4}$$

A lower bound for the right side supplies no lower bound for the left side. R-075 graph recovery closes the regular class in its declared fixed-cutoff Cameron–Martin/terminal-L^6 graph norm; it does not reverse (7.4).

The required new gate can be closed in either of two ways:

1. extend the complete endpoint lower bound to every finite-energy progressive covariance-flow control, including within-shell adaptation, same-range revisits, temporal refinement, auxiliary randomness, localization, and lower semicontinuity, with no new counterterm; or

2. prove the equivalent all-tilted-law entropy inequality directly.

The finite-L^6 restriction also needs a variational truncation showing that finite-objective controls can be approximated without an $\infty - \infty$ defect.

8 Exact conditional Nelson chain

The present state can now be stated without a hidden implication.

Conditional Theorem 8.1. Assume the following three new inputs, uniformly in the cutoff:

(FAR) a bound for (4.4), or the localized gain (4.8), after complete R-063/R-070 reconstruction;

(NEAR) a direct signed near-root bound or a genuine $\gamma > 0$ lower-chaos-complete adapted coefficient estimate for Section 6;

(PROG) the full progressive/revisit extension of Section 7.

Then the R-079 identity, Theorems 3.1–3.2, R-078 paid orientations, and R-066/R-070 trace payments yield the full progressive controlled-shell one-use estimate. With

$$\epsilon_6 = 0.15, \qquad \epsilon_v = 0.45, \qquad p = \frac{11}{10}, \tag{8.1}$$

the production potential allocation has strict sextic margin

$$0.27 - 0.06 - 0.15 = 0.06, \tag{8.2}$$

and the control margin is

$$\frac{1}{2p} - \epsilon_v = \frac{5}{11} - \frac{9}{20} = \frac{1}{220}. \tag{8.3}$$

Exact Boue–Dupuis then gives

$$q = \frac{1}{2\epsilon_v} = \frac{10}{9}, \qquad q - p = \frac{1}{90} > 0, \tag{8.4}$$

and hence

$$\sup_J \mathbb{E} \exp\left[-\frac{10}{9}\{V_J^{\mathrm{ren}} + 0.15\|\phi_J\|_6^6\}\right] < \infty. \tag{8.5}$$

The remaining 0.06 production sextic margin and $q > p$ give the A7 $p = 1.1$ Nelson bound. The already proved A7 measure-theoretic implications then yield the fixed-volume full-sequence Gibbs limit and smeared-correlation convergence at the fixed floor. ☐

The theorem is conditional because FAR, NEAR, and PROG are all open. In particular, (8.5) is not a conclusion of this note.

9 Evidence map, failures, and remaining work

The proof map after R-080 is:

$$
\begin{array}{c}
\text{R-079 exact complete packet}\\
\downarrow\\
\boxed{\text{LOW-1 closed}}\quad \boxed{\text{LOW-2 closed}}\quad \boxed{\text{FAR reduced to } S_C}\quad \boxed{\text{NEAR narrowed}}\\
\downarrow\quad \text{FAR+NEAR still required}\\
\text{regular packet lower bound}\\
\downarrow\quad \text{PROG still required}\\
\text{full progressive one-use}\\
\downarrow\quad \text{exact arithmetic already checked}\\
q = 10/9 \text{ Nelson and fixed-floor A7 construction.}
\end{array} \tag{9.1}
$$

Successful steps and failed routes are deliberately recorded together:

- **Closed:** both low objects in the regular no-revisit class.
- **Closed reduction:** far feedback is absorbed by orthogonal square completion; only the predictable base-current tail remains.
- **Narrowed:** the residual near explicit payload is predictable; the hidden adapted coefficient is not.
- **Failed route:** target-space heat projection as an automatic spatial tail, by (5.1)–(5.4).
- **Failed route:** bounded-width near localization as a source of Young slack; its slack is zero.
- **Failed route:** universal near-root PSD/Ward positivity, by (6.5).
- **Failed implication:** regular graph recovery to full progressive Boue–Dupuis control, by (7.3)–(7.4).

For the fixed-floor A13/A7 branch, three substantive analytic inputs remain: FAR, NEAR, and PROG. FAR and NEAR may ultimately be proved by one common production paracomposition theorem, but no such theorem currently exists. After those inputs, budget allocation and the Nelson/A7 synthesis are moderate bookkeeping rather than a fourth unknown mechanism. For all of Sector A, the separate A6 full-field bare-concentration branch remains a major independent block; floor removal and infinite volume remain outside the present theorem scope. A descriptive estimate is that roughly 85–90 percent of the A13 algebraic architecture is mapped, but the remaining bridges are the hardest analytic ones, so that percentage is not a time-to-completion forecast.

10 Devil's-advocate review

1. **Objection: "Dropping PSD terms in (3.2) silently merges the two low objects." DISMISSED.** Theorem 3.1 uses only (2.2). Theorem 3.2 separately starts from (2.3) and invokes R-066 plus the exact R-078 paid terms.
2. **Objection: "The low estimates are uniform over every progressive drift." UPHELD as false with mitigation.** The scope is explicitly no-revisit. Equations (7.1)–(7.3) give a quartic termwise counterfixture outside that class, and the new PROG gate is registered.
3. **Objection: "R-075 density already supplies PROG." UPHELD as false.** R-075 closes a specified graph of orthogonal one-shot controls. It neither generates same-range revisits nor changes the infimum direction (7.4).
4. **Objection: "Far square completion proves the far bound." UPHELD as false.** It reduces the feedback loss to S_C in (4.4). The localized predictable base-current tail remains a new theorem.
5. **Objection: "The heat semigroup gives the missing spatial decay." UPHELD as false.** It acts in target space. The third harmonic in (5.1) and the strict-past gap saturation (5.3) survive.
6. **Objection: "Near-root bounded width is automatically subcritical." UPHELD as false.** At $\gamma = 0$ the energy and sextic powers sum to one. A genuine gain or signed production cancellation is necessary.
7. **Objection: "The negative fixture (6.5) disproves one-use." DISMISSED.** It disproves universal rootwise positivity only. A frequency-local budgeted signed estimate may couple roots and shells.
8. **Objection: "The arithmetic (8.2)–(8.4) proves Nelson." UPHELD as an overclaim.** It proves only that the budgets close once FAR, NEAR, and PROG have been supplied.

11 Reproduction and independent audit

The primary executable checks the PSD low-current drop, optimal scalar Young fixtures, far-shell square completion, root-gap saturation, target-heat harmonics, near gain ledgers, the production sign fixture, revisit quartic scaling, variational direction, and the exact Nelson margins. The non-importing independent executable uses complex Hilbert coordinates, a direct discrete Fourier transform, a separate spatial grid, alternate parameters, and separately derived rational ledgers.

Run from the repository root:

```
python codes/foundations/
  a13_classii_low_object_far_square_progressive_boundary_verify.py
```

The child contracts are primary 51/51 and non-importing independent 35/35. The integrated verifier pins R-066, R-075, R-078, and R-079; checks sources, note, PDF, manifest, run, result, negative-result, gate, roadmap, task, and claim surfaces; and fails closed if any open production, progressive, one-use, Nelson, measure, or Sector-A flag is promoted.

Result footer

```
Result ID: A13-CLASSII-LOW-OBJECT-FAR-SQUARE-PROGRESSIVE-
           BOUNDARY
Precise statement:       For regular mutually orthogonal one-shot shell
                         controls with no low-range revisit, both the R-079
```

	conditional low current and complete low endpoint are absorbed below arbitrary terminal-sextic budget. Orthogonal far-shell completion reduces all future feedback to the predictable base-current tail S_C. The near explicit payload is predictable, but its hidden future coefficient remains. R-075 does not extend these results to all progressive controls.
Scope:	Finite cutoff; fixed floor; common even regulator; exact products; strict-past one-shot whole shells; no later low-range revisit; constants J-uniform.
Dependencies:	A1; R-050; R-063; R-066; R-070; R-075; R-078; R-079.
Evidence grade:	ANALYTIC, EXACT, EXECUTED.
Reproduction command:	python codes/foundations/ a13_classii_low_object_far_square_progressive_ boundary_verify.py
Expected output:	Primary 51/51; independent 35/35; manifest-pinned integrated and aggregate PASS; exit zero.
Falsification gate:	Failure of either low theorem, far completion, root-gap/revisit fixture, rational ledger, evidence hash, PDF contract, or honesty flag.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	No production far tail, near signed/gain theorem, full-progressive extension, controlled-shell one-use, Nelson bound, measure theorem, removal limit, or Sector-A closure is proved.
Next required action:	Prove FAR and NEAR for the complete regular packet, then prove the full progressive/revisit extension; only then invoke the q=10/9 conditional synthesis.